

# Theory Of Automata

## Sessional-II Exam

Date: November 6th 2025

Total Time (Hrs): 01

Course Instructor(s)

Total Marks: 30

Ms. Shaharbano, Ms. Bakhtawar, Ms. Abeeha,

Total Questions: 02

Mr. Kashan, Mr. Kashif, Mr. Ubaid Ullah

Roll No

Section

Student Signature

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### INSTRUCTION:

- Return the question paper.
- Read each question completely before answering it. There are 3 questions on 1 page 2 sides.
- In case of any ambiguity, you may make assumptions. But your assumption should not contradict any statement in the question paper.
- Start each question in a new page.

**CLO #2: Prove properties of languages, grammars and automata with rigorously formal mathematical methods.**

### Question 1a: Kleene's Theorem:

[10 mins, 6 Points]

Find the Concatenation of FA1 and FA2, given in figure 1 and figure 2, and the closure of FA2 given in figure 2.

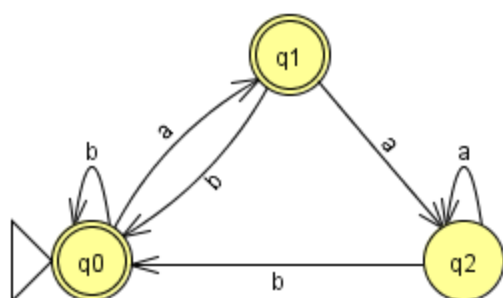


figure 1: FA1

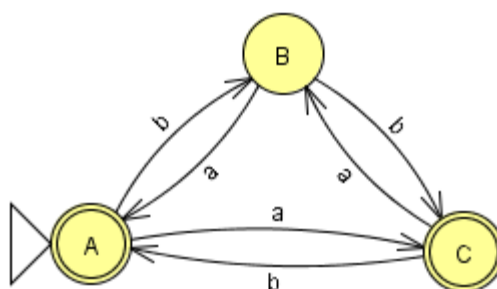


figure 2: FA2

### Question-1b: Grammars:

[20 mins, 3+3+3+3+2 Points]

Given the following Language, identify whether the language is a regular or context-free language, then express the respective grammar. For regular language, create a Regular Grammar, or for CFL create CFG: Convert RG of part 2 into Equivalent FA

1.  $(u^{2n+3})(v^n)$ ;  $n > 0$
2.  $L1 = \{w \in \{a,b\}^* \mid w \text{ starts and ends with the same symbol}\}$
3.  $L2 = \{w \mid |w|_a = 3 \bmod 5\}$ , where  $\Sigma = \{a,b\}$

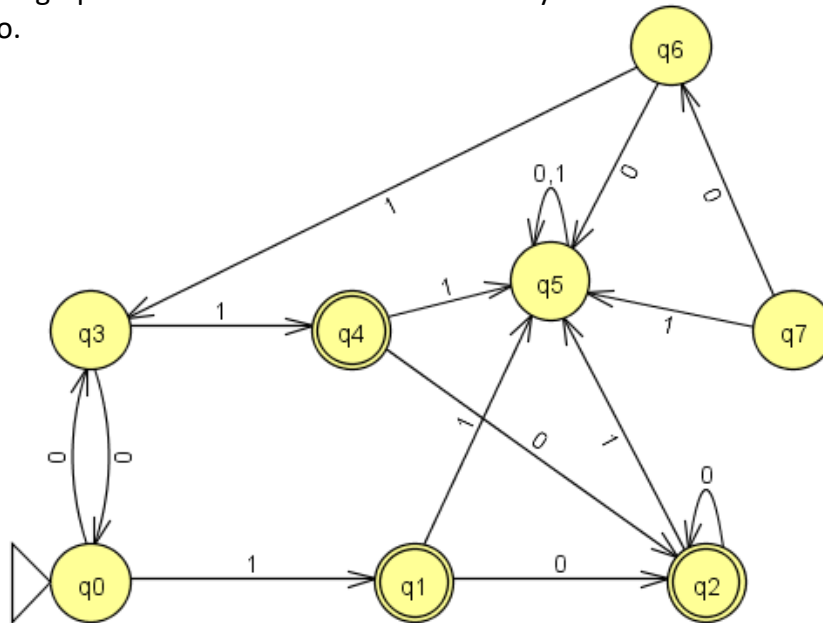
For example: aaa, aabbbaabbbbaabbbba, ababababababababababaaa

4. All strings with exactly two a's and more than two b's.
5.  $L = \{a^n b^m c^k \mid n = m \text{ or } m \leq k\}$

**Question 2a: DFA Minimization:**

**[30 mins, 5+5 Points]**

Given the following DFA M. apply the DFA minimization algorithm and show the resulting minimized DFA. Show your graph or table and label each state in your minimal DFA with the set of states of M it corresponds to.



**Question-2b Simplification:**

$S \rightarrow ASA \mid AB \mid BCD \mid DFG \mid EFG \mid AFG$

$A \rightarrow Abc \mid BCa \mid aAc \mid \epsilon$

$B \rightarrow BD \mid AD \mid bAD$

$C \rightarrow cCb \mid bCD \mid bAC$

$D \rightarrow dD \mid \epsilon$

$F \rightarrow EF \mid CD \mid bCF \mid ef$

Simplify the grammar and show all the steps.

===== Best of Luck =====